

Implicit Video Reasoning for Multiple-Choice Video QA

A Literature Review and Candidate Selection for the ImplicitQA / VRR-QA Challenge

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Abstract

This review surveys the problem, the benchmark, and the modeling landscape for the *ImplicitQA* video question-answering challenge (arXiv:2506.21742, retitled *VRR-QA* in its v3 revision — the namesake of this project). The task requires answering multiple-choice questions whose answers are *not* directly observable in any single frame but must be *inferred* from spatial layout, motion, depth, viewpoint, causality, and social context across discontinuous frames of creative video. We characterise what makes the dataset distinctive — a small, expert-curated, reasoning-first benchmark with an enormous human-machine gap — grounding the analysis in the exact statistics of the local release (**ImplicitQAv0.1.2**). We then go beyond the original baselines to survey the *latest* (2025–2026) methods: reinforcement-learning / GRPO video reasoners (Video-R1, VideoChat-R1/R1.5, DeepVideo-R1), R1-Zero-style visual-spatial RL (vsGRPO), iterative-perception test-time scaling, and analog reasoning benchmarks (Video-Holmes). Every quantitative claim in Sections 3–6 was checked by a 3-vote adversarial verification pass; surviving caveats are reported explicitly. We close with a ranked, vLLM-feasible set of candidate models and inference strategies.

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1 The task and the competition

The challenge targets **implicit video reasoning**: answering questions that “cannot be answered through direct observation of frames alone” and that demand “reasoning about off-screen events, unstated motives, or causal chains spanning multiple clips” [1]. This is a deliberate departure from mainstream VideoQA benchmarks (Video-MME, MVBench, MLVU), which are dominated by *explicit*, perception-style questions answerable from a single salient frame. The task is **multiple choice** (typically 4 options, 2–7 in practice), scored by accuracy and macro-averaged accuracy across reasoning categories. The project pipeline mirrors this: it serves a vision–language model (VLM) through vLLM, samples frames per clip, prompts for chain-of-thought (CoT) followed by an **Answer:** `<letter>` line, parses the letter robustly, and optionally aggregates several sampled reasoning paths by majority vote (self-consistency).

2 What is special about the dataset

2.1 Headline properties

ImplicitQA/VRR-QA [1] contributes $\approx 1,000$ expert-annotated QA pairs over $\approx 1,000$ high-quality creative video clips spanning 15 genres across 7 decades (mixed animated and live-action). Four properties make it unusual and should drive every modeling decision:

1. **Reasoning-first, not perception-first.** Every question is constructed so the answer is *not* literally on screen. Frame captioning, OCR, or single-frame grounding — the workhorses of most VideoQA systems — are largely insufficient.
2. **Small and clean.** At $\sim 1\text{k}$ items it is an *evaluation* benchmark, not a training set. There is no in-distribution data to fine-tune on; gains must come from better base models, prompting, and test-time compute.
3. **Large human–machine gap.** Non-expert humans reach **83.0%** overall / 85.6% macro, while the best proprietary *reasoning* model (GPT-o3) reaches only $\approx 64.1\%$ / 68.6%, GPT-4.1 only 54.3%, and *no open-source model crosses 50% overall* (InternVL3 reaches 50.2% only on the category-balanced *macro* metric) [1]. The headroom is enormous.
4. **Spatial reasoning dominates.** Four of the nine categories (Lateral, Vertical, Relative Depth/Proximity, Viewpoint/Visibility) are spatial, and in the local release they account for $\approx 69\%$ of all questions — a known weakness of current VLMs and the single most leverageable axis.

2.2 Category taxonomy and local distribution

Table 1 reports the exact counts in the local `ImplicitQAv0.1.2.jsonl` release (1,001 QA pairs over 327 unique videos). *The spatial categories alone are the majority of the benchmark.*

2.3 Structural quirks that affect scoring and prompting

Direct inspection of the local file surfaces several practically important facts:

Category	# QA	Share	Type
Relative Depth and Proximity	266	26.6%	spatial
Vertical Spatial Reasoning	220	22.0%	spatial
Lateral Spatial Reasoning	161	16.1%	spatial
Motion and Trajectory Dynamics	91	9.1%	temporal
Causal and Motivational Reasoning	82	8.2%	causal
Inferred Counting	58	5.8%	aggregative
Physical and Environmental Context	53	5.3%	context
Viewpoint and Visibility	41	4.1%	spatial
Social Interaction and Relationships	29	2.9%	social
Total	1001	100%	

Table 1: Category distribution in the local release. The four spatial categories (depth + vertical + lateral + viewpoint) sum to $\approx 68.8\%$ of the benchmark.

- **Variable option counts.** Not all questions are 4-way: 4 options ($n=613$), 3 (252), 2 (102), 5 (27), 6 (5), 7 (2). A fixed “A–D” assumption is wrong; the parser and the random baseline must be option-count aware. Average chance is therefore *above* 25% (closer to $\sim 30\%$), so a model must clear that bar to add value.
- **Answer-key prior.** Gold letters are imbalanced: B (302) > A (296) > C (238) > D (159), with E (5) and G (1) rare. **B is the majority class**, which is exactly why the pipeline uses B as the fallback choice — a data-driven default for un-parseable outputs or load failures.
- **Short clips, broad range.** Question windows are short on average (mean 20.2 s, median 12.6 s) but range 1.1–162 s. Short median duration means *dense* sampling within a short window is feasible and preferable to sparse long-video sampling; a few long clips justify a modest upper frame budget rather than a uniform low one.

3 Benchmark evidence: what the numbers say

The original benchmarking [1] provides the clearest prior for candidate selection (Table 2, 16 uniformly sampled frames).

System	Overall	Macro
Human (non-expert)	83.0	85.6
GPT-o3 (reasoning)	64.1	68.6
GPT-4.1 (full)	54.3	58.6
Qwen2-VL-7B	44.9	46.0
LLaVA-OneVision-7B	43.4	46.4
InternVL3	43.3	50.2
Qwen2.5-VL	42.8	–
LLaVA-Video-7B	42.1	46.3
Gemma-3	42.1	–

Table 2: Reported ImplicitQA accuracy (%). No open model crosses 50% overall; the reasoning-oriented GPT-o3 leads by a wide margin [1].

Findings that carry directly into our design:

1. **Reasoning > scale.** GPT-o3 beats GPT-4.1 by ≈ 9.8 points purely from reasoning-style inference — a larger jump than any model-size step among non-reasoning models. *Explicit step-by-step reasoning is the highest-yield lever*, which is precisely what the RL methods in Section 5 try to instill into small open models.
2. **Structured prompts help measurably.** Structured reasoning prompts lifted GPT-4.1 mini, with vertical spatial reasoning improving notably. This validates the project’s CoT prompt and motivates *category-aware* variants.
3. **Frames saturate early.** 8→16 frames helps modestly; beyond 16, accuracy saturates (Qwen2.5-VL gained $\sim 2\%$ from 16→32). Given short median clips, **16–32 frames** is the sweet spot; spend compute on reasoning.
4. **Counting and depth are hardest** for every model — the categories where targeted prompting / aggregation can most differentiate.

4 Latest methods (2025–2026)

The most important development since the original paper is the arrival of **reinforcement-learning (RL) post-training for video reasoning** — the open-source analog of the “reasoning > scale” effect that lets GPT-o3 lead the benchmark. All methods below are built on the Qwen-VL family and are therefore **drop-in compatible with the project’s vLLM pipeline**.

4.1 RL / GRPO video reasoners

Video-R1 [2] (NeurIPS 2025). The first systematic R1 paradigm for video reasoning. It introduces **T-GRPO** (temporal GRPO), whose contrastive ordered-vs-shuffled-frame reward explicitly forces the model to use temporal information rather than reason from a single frame — directly relevant to ImplicitQA’s cross-frame questions. Built on **Qwen2.5-VL-7B**. Reported: **35.8–37.1%** on VSI-Bench (frame-count dependent), *surpassing GPT-4o* (34.0%), and **36.5%** on Video-Holmes, beating o1-mini (29.9%) and Gemini-2.0-Flash (30.6%). **Fully open** (MIT): paper, code, 7B weights, and two datasets (**Video-R1-CoT-165k** for SFT cold-start, **Video-R1-260k** for RL); an intermediate SFT checkpoint lets you skip SFT. As a Qwen2.5-VL-7B derivative it runs unchanged in vLLM. *This is the single best-evidenced open candidate.*

VideoChat-R1 [3] (NeurIPS 2025). Rule-based **spatio-temporal Reinforcement Fine-Tuning (RFT)** as a data-efficient post-training method, with large task-specific gains (e.g. +31.8 temporal grounding, +31.2 object tracking) without sacrificing base capability. Open code and multiple HF checkpoints (**VideoChat-R1_7B**, **-thinking**, **-caption**) from OpenGVLab; Qwen2.5-VL-based, vLLM-feasible.

VideoChat-R1.5 [4] (NeurIPS 2025). Adds **Visual Test-Time Scaling (VTTS)** via an **Iterative Perception (ITP)** loop: RL-trained spatio-temporal supervision progressively refines focus on high-confidence regions guided by updated text predictions — a “think-then-look” agentic loop well-suited to off-screen/implicit reasoning. Built on Qwen2.5-VL 3B/7B, open-source. Gains are real but modest (see §5.3).

DeepVideo-R1 [5] (NeurIPS 2025). Trains with **Reg-GRPO** (regressive GRPO, reformulating the GRPO loss as direct advantage regression and removing clipping/min safeguards) plus difficulty-aware data augmentation, fixing two GRPO failure modes specific to video LLMs (safeguard reliance, vanishing advantage). Open code; reports gains across SEED-Bench-R1, LongVideoBench, NExT-GQA.

4.2 Visual-spatial RL (the highest-relevance axis)

Since $\approx 69\%$ of the local benchmark is spatial, methods that specifically boost visual-spatial reasoning are unusually relevant. **vsGRPO** (“Improved Visual-Spatial Reasoning via R1-Zero-Like Training”) [6] applies GRPO-based, R1-Zero-style RL on a curated **VSI-100k** dataset. Reported: **vsGRPO-2B** (from Qwen2-VL-2B) beats its base by 12.1% and *surpasses GPT-4o* on VSI-Bench; **vsGRPO-7B** matches the best open model LLaVA-NeXT-Video-72B (≈ 40.9). Caveat: built on the older *Qwen2-VL* (not 2.5), self-reported on VSI-Bench only. The *recipe* (spatial-data GRPO) is the transferable lesson even if the released checkpoint is not the best base.

4.3 Test-time scaling, keyframes, self-consistency

Test-time perception scaling yields *real but modest* gains: VideoChat-R1.5’s iterative perception improves VideoMME 65.2 $\rightarrow$ 67.9 (+2.7) and LongVideoBench 61.4 $\rightarrow$ 62.9 (+1.5) as perception steps increase, but only +1.4 on VSI-Bench (40.6 vs 39.2) [4]. The largest gains are on *grounding*, not spatial *reasoning*. Implication: iterative perception and self-consistency help, but spatial-reasoning improvements are small, so they should be *combined* (RL base + frame strategy + voting) rather than relied on singly. Question-conditioned keyframe selection (relevance + coverage) remains a sensible next experiment for the long-clip tail; for the short-median clips here, dense uniform sampling within the question window is a strong default.

4.4 Analog benchmark: where even frontier models fail

Video-Holmes [7] is a close analog: 1,837 MCQ over 270 annotated suspense films, requiring models to *locate and connect multiple scattered clues across segments*. Even **Gemini-2.5-Pro tops out at only 45%**, with most models below 40% — corroborating that reasoning-heavy, cross-frame video QA is unsolved even for proprietary upper bounds, and that this is a genuine research frontier rather than an engineering-only gap.

5 Candidate selection and recommendations

Synthesising the above, Table 3 ranks concrete candidates by expected value under the project’s existing vLLM pipeline.

Recommended program of work

1. **Baseline-to-beat:** run current Qwen2.5-VL-72B-AWQ at 16 frames on **val**; record per-category accuracy.
2. **Swap in an RL reasoner:** register **Video-R1-7B** (then **VideoChat-R1.5-7B**) as presets; compare overall *and* macro. Expect the 7B RL models to be competitive with or beat the 72B non-RL base on reasoning.

3. **Add test-time compute:** CoT + self-consistency $n=5$ on the best base; verify the gain exceeds the cost.
4. **Ensemble** the best two families with the existing vote script; tune weights on `val` (optimise macro for the small categories).
5. **Targeted prompting** for the weak, large spatial categories (depth, vertical, counting) using “describe spatial layout first” prompts.
6. **Frame / keyframe study** only if (3)–(5) plateau: 16 vs 32 frames, then question-conditioned selection on the long-clip subset.
7. **(Optional) spatial GRPO:** if compute allows, probe the vsGRPO recipe on a Qwen2.5-VL base using public spatial data — highest-relevance to the dominant category.

6 Caveats and open questions

This is a fast-moving field; all primary sources are March 2025–2026. The findings above passed 3-vote adversarial verification (24/25 confirmed), but several caveats bound their use:

- **No ImplicitQA-specific results exist beyond the original paper.** All RL/spatial numbers (Video-R1, vsGRPO, etc.) are reported on *VSI-Bench* / *Video-Holmes*, not on ImplicitQA. Transfer is plausible (shared spatial / cross-frame skills) but unproven — *validate every candidate on val before trusting it*.
- **RL gains are small and often self-reported.** Margins over GPT-4o are typically +1.4 to +3.1 points on a single benchmark, with no independent reproduction located. Treat headline “beats GPT-4o” claims as suggestive, not decisive.
- **Base-model mismatch.** Video-R1 / VideoChat-R1 use Qwen2.5-VL; vsGRPO uses the older Qwen2-VL. This affects vLLM compatibility and expected transfer — prefer the 2.5-VL-based checkpoints.
- **Unverified items.** No primary-source video-reasoning numbers for Qwen3-VL, InternVL3.5, or GPT-5-class models survived verification; their entries here are landscape context, not benchmarked claims. A claimed “SpatialStack-5B, VSI-Bench 67.5, rank-1” was **refuted (0–3 votes) and must not be used**. The Dual-Clue Reasoning paper (arXiv:2506.07811) was requested but yielded no verified, runnable method — treat as a lead, not a recommendation.
- **Metric definition matters.** “No open model >50%” holds for *overall* accuracy; InternVL3 reaches 50.2% on the category-balanced *macro* metric. Always report both.

7 Conclusion

ImplicitQA/VRR-QA is a small, clean, reasoning-first benchmark dominated by 3D spatial questions, with an enormous human–machine gap and a clear published signal that *reasoning-style inference matters more than raw scale*. The 2025–2026 literature operationalises that signal for open models via RL / GRPO post-training. The evidence points to a concrete plan: adopt an **RL-tuned Qwen2.5-VL-7B reasoner** (Video-R1, then VideoChat-R1.5) as the base, amplify it with **CoT + self-consistency** and iterative perception, **ensemble** across families, and concentrate prompting on the large, hard *spatial* categories — all within the project’s existing vLLM pipeline. Frame budget should stay modest (16–32); compute is better spent on reasoning than on more frames.

Crucially, because no method has published ImplicitQA-specific numbers, every candidate must be re-validated on the local `val` split before selection.

References

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Candidate	Rationale	Priority
Video-R1-7B (Qwen2.5-VL-7B + T-GRPO)	Best-evidenced open reasoner; temporal reward targets cross-frame reasoning; beats GPT-4o on VSI-Bench; fully open, MIT, drops into vLLM unchanged.	P0
VideoChat-R1.5-7B (RFT + iterative perception)	Spatio-temporal RL plus a think-then-look test-time loop; partly decorrelated from Video-R1 for ensembling.	P0
CoT + self-consistency ($n=5$, $T \approx 0.6$)	Highest-yield test-time lever; reasoning > scale on this benchmark.	P0
Qwen3-VL / InternVL3.5 (largest that fits, AWQ)	Strongest <i>non-RL</i> bases; improved spatial perception; natural upgrade from existing presets and the non-RL reference point.	P1
vsGRPO recipe (spatial-data GRPO)	Directly targets the $\approx 69\%$ spatial slice; surpassed GPT-4o at 2B. Use the released ckpt as a probe; recipe is the transferable win.	P1
Video-R1 + VideoChat-R1.5 vote ensemble	Captures macro-accuracy upside on small/hard categories; <code>ensemble.py</code> already exists.	P1
16–32 uniform frames, max side 448	Empirically saturating budget; matches short median clip length.	P1
Spatial-aware structured prompts	“Describe layout, then reason”; structured prompts lifted spatial categories in the paper.	P1
Question-conditioned keyframe selection	Next experiment if uniform plateaus, esp. for the long-clip tail.	P2
DeepVideo-R1 (Reg-GRPO)	Alternative RL base if Video-R1 / VideoChat plateau; open.	P2
Qwen2.5-VL-72B-AWQ (current)	Solid, already-validated baseline to beat; reference for ablations.	baseline

Table 3: Ranked candidate models and strategies. P0 = do first; P2 = if budget allows. RL-tuned 7B open reasoners now outrank larger non-RL bases on the evidence.